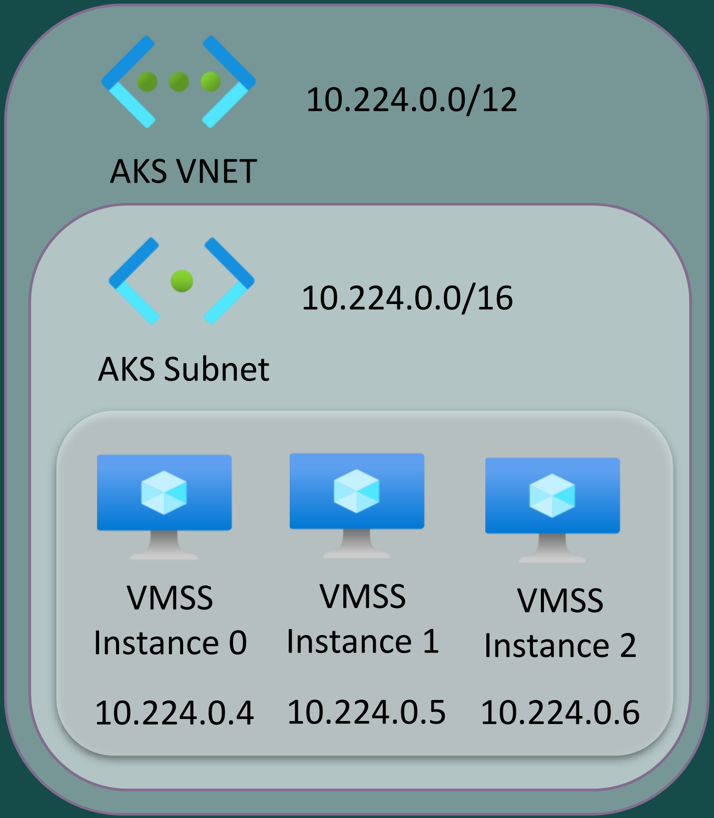
Exploring the AKS cluster – Azure infrastructure side

- **Virtual Network (VNET)**
  - Logically isolated network in Azure that helps to securely connect and manage resources in Cloud
  - You need to specify an address space when creating a VNET
  - The DNS server is set at VNET level
- **Subnet (SNET)**
  - With the help of subnets, you can divide the virtual network into one or more sub-networks and assign each one a specific amount of address space. Following that, you can deploy Azure resources to a specific subnet
  - For every Azure subnet, 5 IPs are reserved by Azure, ending with:
    - .0 (network address)
    - .1 (reserved for default gateway)
    - .2 and .3 (reserved to map the Azure DNS IPs to the VNET space)
    - .255 (broadcast address)



- AKS resources that need an IP will have it from the AKS subnet
- All resources in an AKS cluster need to use the same VNET. It is possible to have node pools in separate subnets, but those subnets need to be part of the same VNET